



TourGuard AI

AI-Based Tourist Verification System



Counts tourists
in starting image



Counts tourists
in current image



Detects missing
tourists



Detects extra
persons



Generates
verification report

PROBLEM STATEMENT

Tour operators and travel agencies frequently face challenges in **verifying whether all tourists have returned** after visiting tourist destinations such as temples, waterfalls, parks, historical monuments, and restaurants.



Currently, tour guides **manually count passengers** at every stop, which can be:



Time-consuming



Error-prone



Stressful for tour guides



Difficult for large groups



Risky if a tourist is accidentally left behind



TourGuard AI solves this problem by automatically counting tourists from group photos using **YOLO11** object detection and comparing the current tourist count with the original tour count.



The system instantly identifies **missing** or **extra persons** and generates a **verification report**.

KEY BENEFITS

- Automated and Accurate Counting
- Reduces Human Errors
- Saves Time and Effort
- Improves Tourist Safety
- Handles Large Groups Efficiently
- Instant Verification Report
- Easy-to-Use Interface

REAL-WORLD APPLICATIONS

- Travel Agencies**
Verify all tourists before leaving any destination.
- Pilgrimage Tours**
Ensure no pilgrims are left behind.
- Educational Tours**
Track students during school and college trips.
- Corporate Tours**
Monitor employee participation in company trips.
- Event Management**
Verify attendees during group outings and events.
- Tourist Safety Monitoring**
Improve safety and reduce the risk of missing persons.

TECHNOLOGIES USED

- YOLO11** (Ultralytics)
- Python**
- OpenCV**
- Gradio**
- Pandas**



Making every journey **safer, smarter** and **stress-free** for tour operators and tourists.



Email:
contact.sugumarai@gmail.com



TourGuard AI

AI-Based Tourist Verification System

Smart Counting. Safe Journeys.
No Tourist Left Behind.



SYSTEM WORKFLOW

- 1 Tour Start Photo**
Capture the initial group photo before starting the tour.
- 2 YOLO11 Person Detection**
Detect and count all persons in the starting photo.
- 3 Expected Tourist Count**
Store the total number of tourists as the expected count.
- 4 Current Group Photo**
Capture the group photo at the current tourist location.
- 5 YOLO11 Person Detection**
Detect and count persons in the current photo.
- 6 Current Tourist Count**
Store the current number of detected persons.
- 7 Compare Counts**
Compare expected and current counts.
- 8 Missing / Extra Detection**
Identify missing tourists or extra persons.
- 9 Alert Generation**
Generate instant alert and verification report.

KEY BENEFITS

- ✓ Automated and Accurate Counting
- ✓ Reduces Human Errors
- ✓ Saves Time and Effort
- ✓ Improves Tourist Safety
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DETECTION OUTPUTS

1. TOUR START PHOTO



2. START GROUP DETECTION (YOLO11 OUTPUT)



3. CURRENT GROUP PHOTO



4. CURRENT GROUP DETECTION (YOLO11 OUTPUT)



TOURGUARD AI REPORT

	Expected Tourists	: 51
	Current Tourists	: 56
	Missing Tourists	: 0
	Extra Persons	: 5
	Status	: ALERT - Extra Persons Detected : 5

TourGuard AI Report



Clear

Submit

Flag

TourGuard AI Report

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Status :
ALERT -
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TECHNOLOGIES USED



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Scan QR to
view project
repository



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